

Syllabus: Probability and Decision

Cian Dorr and Hartry Field

NYU Mind and Language Seminar, Spring 2023

Details

Main session: Tuesdays, 4–7pm. 5 Washington Place, 2nd floor.

Student session: Mondays, 5–6pm. 5 Washington Place, 2nd floor.

Format

Following the tradition of the Mind and Language Seminar, each session (after the first three) will be devoted to discussion of one or more papers by a visitor who will be present for the session. Papers will be pre-circulated about two weeks in advance. Each session will begin with a presentation by at least one of the seminar leaders summarizing and raising questions about the reading; after that, the author will have a chance to respond, and then we will shift to open discussion.

Everyone is welcome to attend the Monday student sessions, which will be devoted to preliminary discussion of the readings and will also attempt to explain any presupposed technical concepts.

Expectations for students taking the course for credit

- Attend the student sessions, and arrive ready to discuss the readings.
- Write a seminar paper (15–20 pages) on a topic relating to probability and/or decision. The due date is subject to negotiation but can in no case be later than the end of the summer.
- Formulate a plan for the paper and discuss it with at least one of the instructors by the beginning of May.

Schedule

Jan 24	<i>Introductory presentation by Hartry</i>
Jan 31	<i>Introductory presentations by Hartry and Cian</i>
Feb 7	<i>Introductory presentation by Cian</i>
Feb 14	Kevin Dorst (MIT)
Feb 21	Wayne Myrvold (Western University)
Feb 28	Jeff Russell (USC)
March 7	Dmitri Gallow (Australian Catholic University)
March 21	Kenny Easwaran (Texas A&M)

March 28	Yoaav Isaacs (Baylor)
April 4	Miriam Shoenfield (University of Texas)
April 11	Julia Staffel (University of Colorado)
April 18	Snow Zhang (NYU) and Alexander Meehan (Yale)
April 25	Lara Buchak (Princeton)
May 2	Catrin Campbell-Moore (Bristol)

Background readings

The seminar is intended to be accessible to those with no prior background knowledge of the philosophical literature on probability and decision; however, given that the schedule of topics after the first three weeks is determined by the visitor's availability and what they are working on, it is likely to be helpful to supplement the pre-circulated readings with further readings of a more introductory character. The following resources may be useful:

- Kenny Easwaran, 'Bayesianism I: Introduction and Arguments in Favor': <https://philpapers.org/rec/EASBII> and 'Bayesianism II: Applications and Criticisms': <https://philpapers.org/rec/EASBIA>. Accessible introductions, published in *Philosophy Compass*.
- Joseph B Kadane, *Principles of Uncertainty*, CRC Press, 2010. A much more mathematically-oriented introduction to Bayesian methods.
- Richard Pettigrew and Jonathan Weisberg (eds.). *The Open Handbook of Formal Epistemology*: <https://philarchive.org/rec/PETTOH-2>. Includes short introductions to Bayesian probability (by Mike Titelbaum) and to decision theory (by Johanna Thoma), as well as to some more specialized topics.
- Michael Titelbaum, *Fundamentals of Bayesian Epistemology I: Introducing Credences* and *Fundamentals of Bayesian Epistemology II: Arguments, Challenges, Alternatives* (both Oxford University Press, 2002). Pretty introductory and oriented to philosophers — check them out if you like Titelbaum's chapter in the Pettigrew and Weisberg volume.
- Stanford Encyclopedia of Philosophy entries on [decision theory](#), [normative expected utility theory](#), [Bayesian epistemology](#), [Bayes' theorem](#), as well as many other relevant more specialized topics.
- Michael Strevens, lecture notes on Bayesian confirmation theory: <http://www.strevens.org/bct/BCT.pdf>. Focuses on applications of the Bayesian formalism in the philosophy of science.
- Brian Weatherson, untitled book manuscript on decision theory and game theory: <http://brian.weatherson.org/424/DTBook.pdf>. Very accessible; seems to be derived from notes for a graduate seminar. Game theory doesn't come in till the last 1/3.